

8 Analytic Geometry and Calculus

8.1 Axes and Co-ordinates

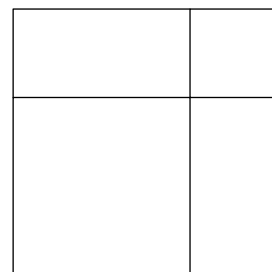
Modern mathematics is dominated by *algebra*. We trust equations and the rules of algebra more than pictorial, geometric reasoning. For example, when modern mathematicians are likely to interpret the expression $(x + y)^2 = x^2 + 2xy + y^2$ within the laws (axioms) of algebra:

$$\begin{aligned}
 (x + y)^2 &= (x + y)(x + y) && \text{(definition of 'square')} \\
 &= x(x + y) + y(x + y) && \text{(distributive law)} \\
 &= x^2 + xy + yx + y^2 && \text{(distributive law twice more)} \\
 &= x^2 + 2xy + y^2 && \text{(commutativity)}
 \end{aligned}$$

For most of mathematical history, however, this result would have been stated *geometrically*, as in Euclid's *Elements* (Thm II. 4):

The square on two parts equals the squares on each part plus twice the rectangle on the parts.

The proof was a simple picture.



We've seen how algebra and algebraic notation were slowly adopted in renaissance Europe. While its utility for efficient calculation was noted, algebra was not initially considered acceptable for the purposes of *proof*; practitioners still felt the need to justify calculations geometrically. In modern times this seems backwards: if a college student today were asked to prove Euclid's result,⁵⁹ they'd likely start by labeling the 'parts' x and y , before appealing to the algebraic formula at the top of the page! Of course, each piece of algebra has a geometric basis:

- Distributivity is the fact that a rectangle on a side and two parts equals the sum of the rectangles on the side and each of the parts respectively.
- Commutativity is the fact that a rectangle has the same area after a 90° rotation.

Modern mathematics has converted geometric rules into algebra and largely forgotten their origin. The slow triumph of algebra over geometry is one of the great revolutions of mathematical history, completely changing the way mathematicians *think*. More practically, algebra permits easy generalization: how would one geometrically justify an expression such as

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4 ?$$

Euclidean Geometry is *synthetic*: based on purely geometric axioms without formulæ or co-ordinates. *Analytic geometry* married algebra, geometry, and number, by introducing *axes* and *co-ordinates*. Modern geometry is primarily analytic and it is now rare to find a mathematician working synthetically—algebra's domination of Euclidean geometry is total. The critical contribution to this revolution was made almost simultaneously by two Frenchmen...

⁵⁹But importantly *not* a 5th-grader, who'd likely consider algebra to be alien and pointless: *isn't the picture obvious?* Keep this in mind if you ever have the privilege of teaching at grade-school: it is *your* ideas that are strange...

Pierre de Fermat (1601–1665)

Fermat made great strides in several areas (probability, analytic geometry, early calculus, number theory, optics), even though mathematics was a pastime rather than his profession.⁶⁰ You've likely encountered his name attached to two famous results in number theory:

Fermat's Little Theorem If p is prime and x an integer, then $x^p \equiv x \pmod{p}$.

Fermat's Last Theorem If $n \in \mathbb{N}_{\geq 3}$, then $x^n + y^n = z^n$ has no integer solutions with $xyz \neq 0$. Fermat is not believed to have proved this beyond a special case ($n = 4$). This problem has captivated mathematicians ever since, motivating several developments in algebra and number theory; a complete proof finally came in the 1990s.

Some of Fermat's fame comes from his enigma. Much of what we know comes from letters to friends in which he claims much but rarely offers complete proofs—it is often unclear whether he merely suspected a general statement. When Fermat died, his notes and letters contained many unproven claims. Leonhard Euler (1707–83) in particular expended much effort on these. Fermat's work on analytic geometry was largely developed by 1629, though it wasn't published during his lifetime. His approach was similar to that of Descartes which we describe below. The two corresponded, not always cordially, but both acknowledged the other's contributions. We'll return to Fermat when we discuss the beginnings of calculus in the next section.

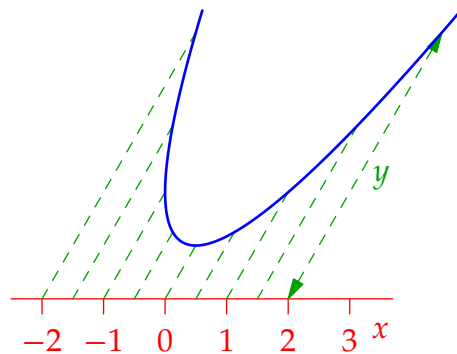
René Descartes (1596–1650)

The chalk to Fermat's cheese, Descartes rigorously recorded everything. His defining work is 1637's *Discours de la méthode*...⁶¹ While enormously influential in philosophy, *Discours* was intended to lay the groundwork for investigation within mathematics and the sciences—Descartes finishes *Discours* by commenting on the necessity of experimentation in science and on his reluctance to publish due to his fear of suffering the same persecution as Galileo.⁶² The copious appendices to *Discours* contain Descartes' scientific work. It is in one of these, *La Géométrie*, that Descartes introduces axes and co-ordinates.

We now think of Cartesian axes and co-ordinates as *plural*, but both Fermat and Descartes used only one axis. Here is a sketch of their approach.

Draw a straight line (the **axis**) containing two fixed points labelled 0 (the *origin*) and 1. All points on the axis are identified with numbers x (originally only positive).

A **curve** is described as an algebraic relationship between x and the **distance** y from the axis to the curve measured using a family of **parallel lines**.



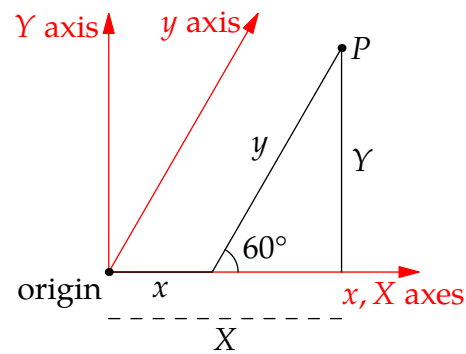
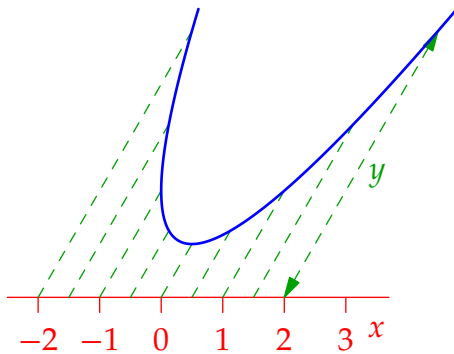
⁶⁰Fermat was wealthy but not aristocratic, attending the University of Orléans for three years where he trained as a lawyer.

⁶¹...of rightly conducting one's reason and of seeking truth in the sciences. The primary part of this work is philosophical and contains his famous phrase *cogito ergo sum* (I think therefore I am).

⁶²At this time, France was still Catholic. Descartes had moved thence to Holland in part to pursue his work more freely. In 1649 Descartes moved to Sweden where he died the following year.

While neither Descartes nor Fermat had a second axis, their approach implicitly imagines one, namely the **measuring line** through the origin. It therefore makes sense for us to speak of the *co-ordinates* (x, y) ; the modern terms *abscissa* (x) and *ordinate* (y) date from shortly after Descartes. It wasn't long before a second axis orthogonal to the first was instituted (Frans van Schooten, 1649), an approach that quickly became standard.

Example 1 The previous picture shows some of the flexibility inherent in Descartes' approach. The curve $y = x^2 + 1$ is drawn, where the 'y-axis' is inclined 60° to the horizontal. To represent the curve in a more standard fashion, we perform a change of co-ordinates. Suppose a point P has co-ordinates (x, y) with respect to the original slanted axes and (X, Y) with respect to the familiar orthogonal axes of van Schooten.



The second picture shows the relationship between the two co-ordinate systems

$$\begin{cases} X = x + y \cos 60^\circ = x + \frac{1}{2}y \\ Y = y \sin 60^\circ = \frac{\sqrt{3}}{2}y \end{cases}$$

A little algebra quickly shows that $\sqrt{3}X - Y = \sqrt{3}x$. It follows that for any point on the curve, we have

$$\begin{aligned} (\sqrt{3}X - Y)^2 &= 3x^2 = 3(y - 1) = 3\left(\frac{2}{\sqrt{3}}Y - 1\right) \\ \Rightarrow 3X^2 - 2\sqrt{3}XY + Y^2 - 2\sqrt{3}Y + 3 &= 0 \end{aligned}$$

This is an implicit equation for a parabola in rectangular co-ordinates. Descartes has essentially chosen a second axis in such a way as to make the equation of this parabola very simple.⁶³

⁶³That the implicit equation is a parabola may be confirmed in modern language by computing its *discriminant*: a non-degenerate quadratic curve $aX^2 + 2bXY + cY^2 + \dots = 0$ is a parabola if and only if $b^2 - ac = 0$. You've possibly encountered this in a linear algebra course, where the standard approach is to *rotate* both axes instead by diagonalizing a symmetric matrix: indeed the discriminant is negative the determinant of the matrix $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ and $(x \ y)A\begin{pmatrix} x \\ y \end{pmatrix} = aX^2 + 2bXY + cY^2 + \dots$

Example 2 Analytic geometry affords easy proofs of many results that are significantly harder in Euclidean geometry. For instance, here is the famous *centroid theorem*.

The three medians⁶⁴ of a triangle meet at a common point (the *centroid*), which moreover lies two-thirds of the way along each median.

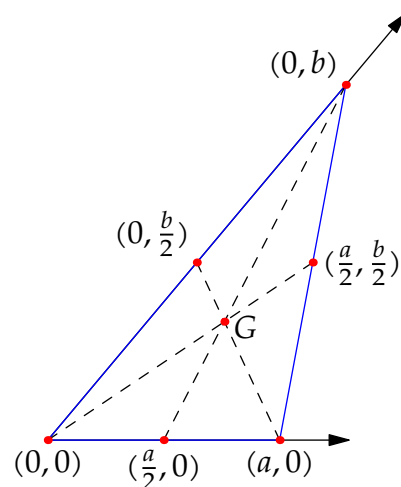
Choose axes pointing along two sides of a triangle with the origin at one vertex. If the two sides lying on the axes have lengths a and b , then the third side necessarily has equation

$$bx + ay = ab$$

Moreover, the midpoints of the three sides plainly have co-ordinates as pictured.

To complete the proof, simply compute the co-ordinates of the point two-thirds of the way along each median: for the median through the upper vertex, this is

$$\frac{2}{3} \left(\frac{a}{2}, 0 \right) + \frac{1}{3} (0, b) = \frac{1}{3} (a, b)$$



One obtains the same co-ordinates for the other two medians, whence *all three meet at a common point G*, the *centroid* of the triangle.

As demonstrated by the centroid proof, Descartes' ability to *choose axes to fit the problem* is a critical advantage of analytic geometry, largely dispensing with the tedious consideration of *congruence* in synthetic geometry.

Descartes found that his method could solve many problems in a more efficient manner than using synthetically; in particular, finding complicated intersections of curves. As we'll see in the next section, such arguments were necessary for the development of calculus, and would often rely on his Factor Theorem (pg. 93).

Given the novelty of his approach, Descartes typically gave geometric proofs to back up his algebraic work. However he also saw the future: once several examples were done, he claimed that it was no longer necessary to draw physical lines and provide a geometric argument; *the algebra was the proof*.

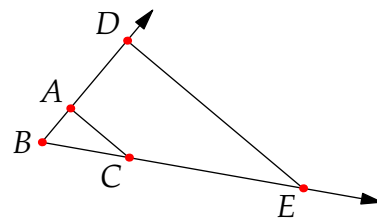
Exercises 8.1. *Key concepts: Basic algebra encodes geometric facts, Analytic geometry helped algebra & number eclipse geometry, Fermat & Descartes' single-axis approach*

1. Assume that $xy = c$ represents a hyperbola with asymptotes the x - and y -axes. Show that $xy + c = rx + sy$ also represents a hyperbola, and find its asymptotes.
2. Determine the locus of the equation $b^2 - 2x^2 = 2xy + y^2$.
(Hint: add x^2 to both sides and remember that axes do not have to be orthogonal...)

⁶⁴A segment joining a vertex and the midpoint of the opposite side.

3. We describe a method whereby Descartes constructed the product of two lengths.

- Let $\overrightarrow{BC}$ and $\overrightarrow{BD}$ be rays forming an acute angle at B . Our goal is to multiply $|BD|$ by $|BC|$.
- Suppose $|AB| = 1$, where A lies on $\overrightarrow{BD}$.
- Join $\overline{AC}$ and draw $\overline{DE}$ parallel to $\overline{AC}$ so that $E \in \overrightarrow{BC}$.

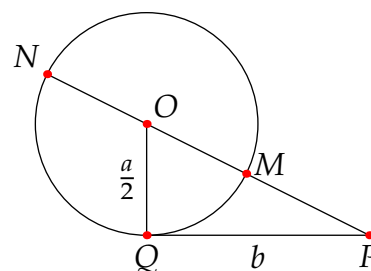


Prove that $|BE|$ is the product of $|BD|$ and $|BC|$.

Similarly, given lengths $|BE|$ and $|BD|$, explain how to construct a segment whose length is the quotient $\frac{|BE|}{|BD|}$.

4. Here is a geometric justification of Descartes for the solution of the equation $x^2 = ax + b^2$, where $a, b > 0$.

- Let $\overline{OQ}$ and $\overline{PQ}$ be perpendicular with lengths $\frac{a}{2}$ and b respectively.
- Draw the circle centered at O with radius $\frac{a}{2}$.
- Draw and extend the line $\overline{OP}$.
- The solution is $x = |NP|$.



- Prove that Descartes' construction indeed recovers a solution.
 - Show that the other solution to the equation $x^2 = ax + b^2$ is negative (*false* to Descartes). How is it visible in the picture?
 - Explain how the same picture could be used to solve the equation $x^2 + ax = b^2$.
5. Following the work of Islamic mathematicians such as Omar Khayyam, Fermat could describe the solutions to certain cubic equations as the intersection of two conics.

For example, to solve $x^3 + bx^2 = bc$ ($b, c > 0$), Fermat would introduce a new variable y by setting both sides of the equation equal to bxy . The (positive) solution to the cubic is the x -solution to the system of equations

$$\begin{cases} x^2 + bx = by \\ xy = c \end{cases}$$

Sketch these curves explicitly for the cubic $x^3 + 4x^2 = 24$. What is the solution?